Day 10: Linux Files and Permissions

Muhammad Usman
Cybersecurity Learning Log

Part 1: Theory

1. What is a Linux directory?
A Linux directory is a location used to organize files and other directories. It works similarly to a folder in other operating systems.

2. What does mkdir do?
The mkdir command creates a new directory. For example, mkdir day10_lab creates a directory named day10_lab.

3. What does touch do?
The touch command creates an empty file if it does not exist. If the file already exists, it updates its modification timestamp.

4. What does cat do?
The cat command displays a file's contents in the terminal. It can also combine the contents of multiple files.

5. What does cp do?
The cp command copies files or directories from one location to another. For example, cp notes.txt backup.txt creates a copy named backup.txt.

6. What does mv do?
The mv command moves files or directories to another location. It can also rename them.

7. What does rm do?
The rm command removes files. Options such as -r allow it to remove directories and their contents recursively.

8. What does ls -l display?
The ls -l command displays files and directories in a detailed format, including permissions, link count, owner, group, size, modification time, and name.

9. Explain r, w and x permissions.
r (read): Allows reading a file's contents or listing directory entries.
w (write): Allows modifying a file or changing directory entries, subject to other permissions and restrictions.
x (execute): Allows running an executable file or accessing/traversing a directory.

10. Explain user, group and others.
User: The owner of the file.
Group: Users associated with the file's assigned group.
Others: Everyone who is neither the owner nor a member of that group.

11. What does chmod do?
The chmod command changes file or directory permissions. For example, chmod u+x demo.sh adds execute permission for the file's owner.

12. What does chmod 755 represent?
It sets permissions to rwxr-xr-x. The owner can read, write and execute; the group and others can read and execute but cannot write.

13. What does chmod 644 represent?
It sets permissions to rw-r--r--. The owner can read and write, while the group and others can only read.

14. What is file ownership?

File ownership identifies the user and group associated with a file or directory. Ownership helps the system determine how permissions apply.

15. What is the purpose of /etc?

The /etc directory commonly stores system-wide configuration files, including settings for services and installed software.

16. What is the purpose of /home?

The /home directory usually contains personal directories for regular users, where they store their files and user-specific settings.

17. What is commonly stored under /var?

The /var directory contains variable data that changes during system operation, such as logs, caches, mail and certain application data.

18. Why should rm be used carefully?

The rm command can permanently remove files, and recovery may not be easy. Incorrect paths or options can cause important data to be deleted.

19. Why are Linux permissions important?

Permissions control who can read, modify, execute or access files and directories. They help protect private information and prevent unauthorized changes.

20. How can incorrect permissions create a security risk?

Overly permissive settings can allow unauthorized users to read sensitive files, modify important data or execute untrusted scripts. Restrictive permissions can also disrupt legitimate services if required access is denied.

Part 2: Practical Lab (Kali Linux)

The lab was done in the Kali Linux terminal inside a new directory called day10_lab. The screenshots below follow the order of the steps.

Figure 1: Create the Lab Directory (mkdir, cd, pwd)



mkdir ~/day10_lab created the directory, **cd** moved into it, and **pwd** confirmed the path /home/kali-linux/day10_lab.

Figure 2: Create an Empty File (touch, ls -l)



touch notes.txt created an empty file. **ls -l** shows it with a size of 0 bytes and permissions -rw-rw-r--.

Figure 3: Write and Display File Contents (echo, cat)



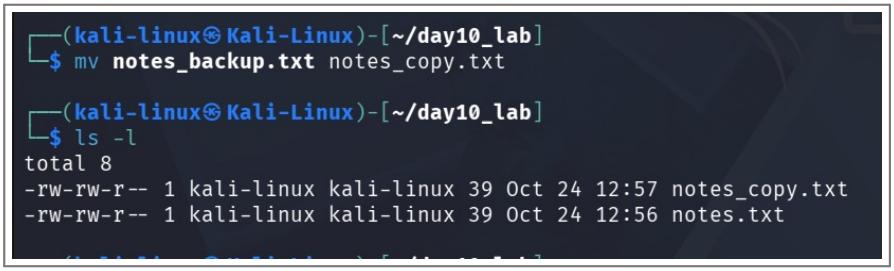
echo with the > operator wrote the text into notes.txt, and **cat notes.txt** displayed it.

Figure 4: Copy a File (cp)



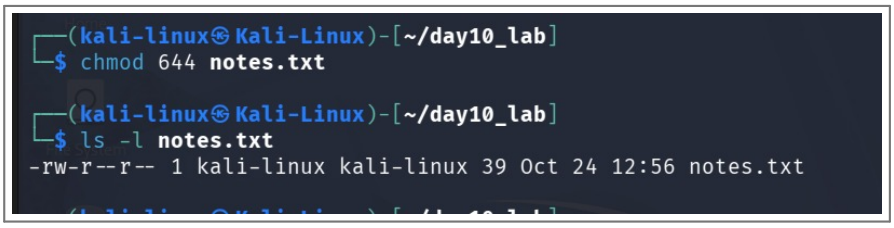
cp notes.txt notes_backup.txt created a copy. Both files are 39 bytes, and **cat** shows the backup holds the same text.

Figure 5: Rename a File (mv)



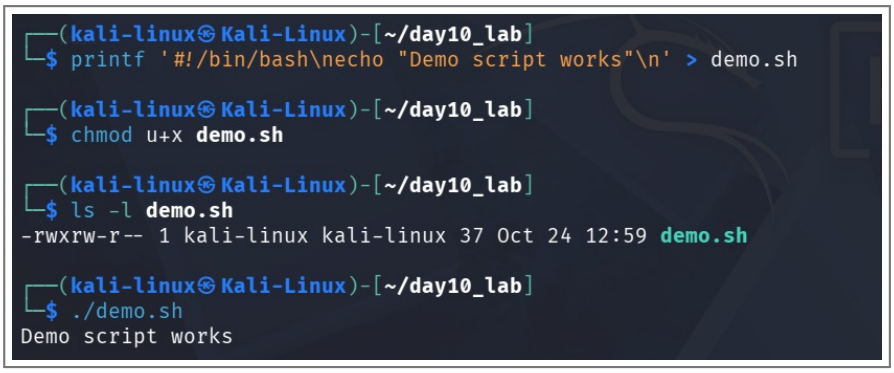
mv notes_backup.txt notes_copy.txt renamed the copy. **ls -l** shows the new name.

Figure 6: Change Permissions (chmod 644)



chmod 644 notes.txt changed the permissions from -rw-rw-r-- to -rw-r--r--. The owner can read and write, while group and others can only read.

Figure 7: Create and Run a Script (chmod u+x)



A small script **demo.sh** was created. **chmod u+x** added execute permission for the owner (-rwxrw-r--), and **./demo.sh** ran it and printed "Demo script works".

Practical Summary

In this lab, I practised creating a directory and files, displaying file contents, copying and renaming files, and changing permissions. These commands are useful for everyday Linux administration and cybersecurity work.